

B.Sc. PHYSICS · DEPARTMENT OF NATURAL SCIENCES · CUEA

Measuring the External Field

A cortical-proxy content observable and the
mean-field bound on media-driven opinion change

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Statistical physics has always modelled media as a **field** pushing on a population of coupled opinions.

That field is assumed. It is not measured.

This project measures one from content itself, applies it to 400 articles, and derives how strong the coupling would have to be for a spread that size to move a population at all.

01

The problem

What is missing from the way media is modelled.

Two articles can report one event and leave readers in different states

WIRE REPORT

Federal Reserve holds interest rates steady, citing a stable inflation outlook.

informs

THE SAME EVENT

FED DESTROYS AMERICA.
Your savings are GONE. The collapse they hid from you!

provokes

Readers notice the difference and cannot quantify it. The platforms distributing both have no objective measure of which they are amplifying.

Existing tools measure the words; none measures the predicted response



A released neural encoder predicts cortical activity from the content, and the measurement is taken on that prediction.

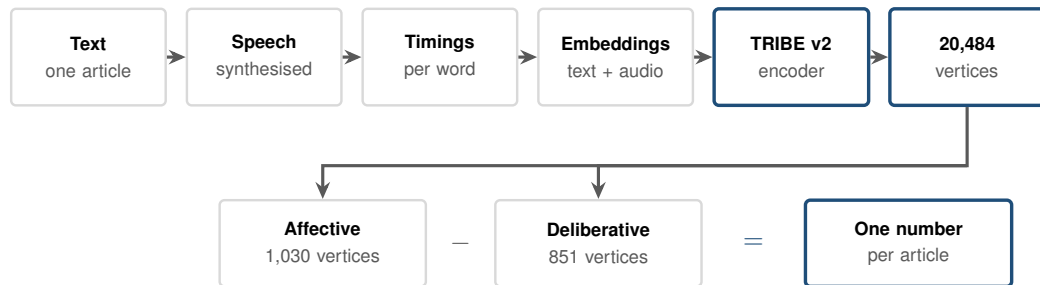
The instrument is a thermometer for one property of media. The thesis is an account of building it, running it, and stating what its readings support.

02

The instrument

From a passage of text to a single number.

A five-stage cascade turns one article into one number



Encoder: TRIBES v2, d'Ascoli et al., Meta FAIR (2025), arXiv:2507.22229. Regions from HCP MMP1.0, Glasser et al. (2016). Instrument at monarch-4iy.pages.dev.

The index contrasts two cortical networks, 1,030 vertices against 851



Affective salience in orange, deliberative control in blue. Resolved once into vertex index sets and cached, so all 400 items are averaged over identical regions.

Regions defined on HCP MMP1.0 (Glasser et al., 2016); surfaces are fsaverage5.

The instrument

The observable is a signed difference, defined for every item

$$NAA_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}$$



A **cortical proxy**: it rates content, never a person, and is never called the amygdala.

Inspecting the checkpoint changed what the project could claim

01

Cortical only

Output is (T , 20484). There is no subcortical head.

The amygdala is not predicted, so the proposal's central equation cannot be computed.

02

Averaged over subjects

The loader forces `average_subjects`.

It predicts a typical viewer. Whether that carries signal is the third paper's question.

03

Standardised output

Region means sit near zero, so a *ratio* sign-flips or explodes.

Undefined for 69 of 400 items. Replaced by the difference.

Each was found by testing the artifact, and each is stated in the abstract rather than the limitations.

03

The measurement

400 articles, scanned twice.

The corpus is length-matched and was powered before the scan

100 fear-activating ISOT-fake	100 high outrage SemEval-2019 T4	100 reward hook Webis-Clickbait-17	100 neutral in- formational PubMed + ISOT-true
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Length-matched. Mean word counts 167.4, 163.5, 163.7, 162.2; standard deviations near 11.

Powered first. Smallest detectable $\eta^2 = 0.0268$, smallest detectable AUC = 0.5916, both fixed before a single item was scanned.

Corpora: Ahmed et al. (ISOT, 2017); Kiesel et al. (SemEval-2019 Task 4); Potthast et al. (Webis-Clickbait-17); PubMed open abstracts.

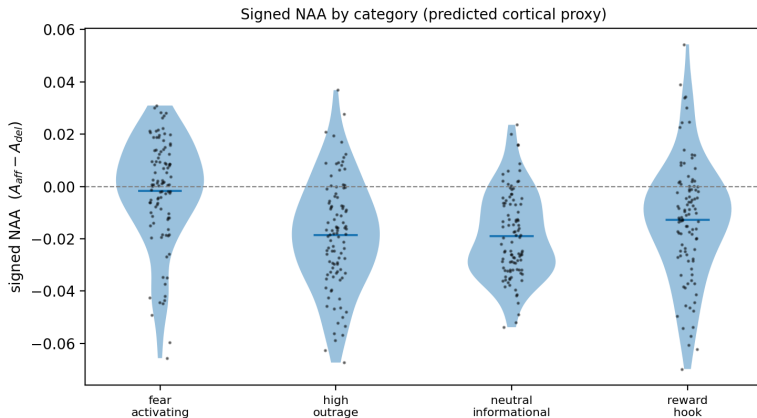
THE FOUR CATEGORIES SEPARATE AT

$$\eta^2 = 0.1068$$

$$F = 15.779, \quad p = 1.03 \times 10^{-9}, \quad n = 400$$

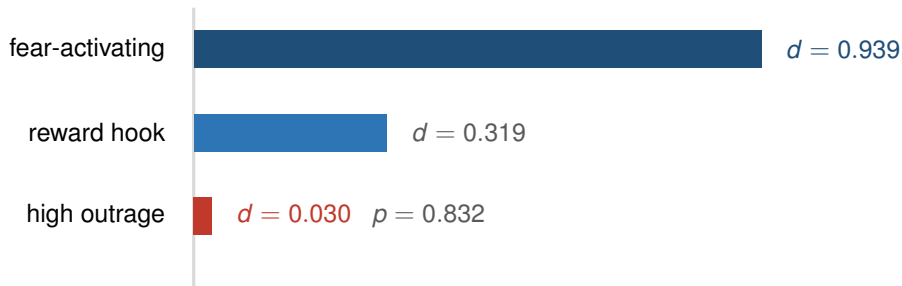
against a design powered to detect 0.0268

Categories shift against each other, but the distributions overlap



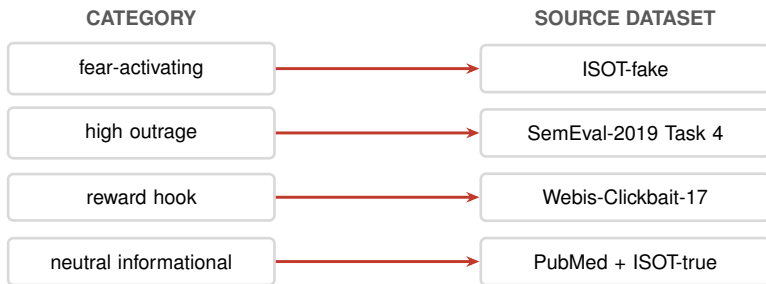
Every category mean is negative: the deliberative network leads throughout, and the categories differ in how far below zero they sit.

Fear-activating content carries the effect; outrage moves both networks equally



Cohen's d against the neutral baseline. **Outrage is the category the proposal expected to separate most**, and on the index it does not. Taken apart, it raises both networks: affective $d = +0.507$, deliberative $d = +0.491$. Fear raises one only.

Category cannot be told apart from source dataset



One source per category, so the separation is between *corpora* and cannot be attributed to *framing*. Length is controlled; provenance is not.

A bag of words beats the index on the label, so the index is not a classifier

0.6274

AUC, the index, nothing fitted

0.9758

AUC, TF-IDF + logistic, 5-fold

Same 400 rows, same pre-scan label. Sentiment (VADER) does not clear chance, 0.5392. On the 150 ISOT items, the one source carrying both classes, TF-IDF still reaches 0.9724. A probe on the same words recovers the source dataset at 66.25% against 25% chance, so much of the lexical win is provenance.

The index is carried forward as the field observable the physics needs, not as a detector.

Group-level claims hold at this precision; per-item claims do not

0.8725

ICC between two GPU sessions

The separation replicates: $\eta^2 = 0.0888$ on the second session against 0.1068 on the first. Both clear the detectable floor.

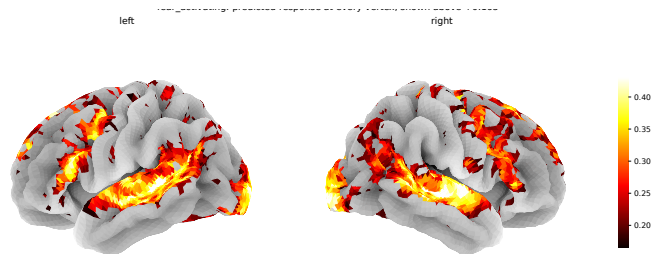
12.8%

of items reverse direction

51 of 400 items flip sign between sessions, so no verdict on one article appears anywhere in the analysis, the figures or the public site.

Additive noise at the measured session error predicts 55.3 reversals, 95% [44, 67]; 51 are observed. Reversing items sit near zero and spread evenly across categories.

The encoder returns a full cortical map, not a summary statistic



One fear-activating item at all 20,484 vertices, painted above the map's 70th percentile.
Nothing is smoothed, interpolated or simulated.

04

The physics

What a measured field is worth.

Media enters the mean-field model as a field on the measured observable

Each person holds an opinion $s_i = \pm 1$, is pulled toward their neighbours with strength J , and is pushed by media with field h :

$$m = \tanh(\beta J m + h), \quad h = \alpha X$$

where X is the measured observable and α the unknown coupling. Expanding $\operatorname{artanh}(m) \approx m + m^3/3$:

$$F(m) = am^2 + bm^4 - hm, \quad a = \frac{1 - \beta J}{2}, \quad b = \frac{1}{12}$$

Coefficients **derived** from the self-consistency condition. The proposal carried $a = 1 - \beta J$, $b = (\beta J)^3/3$, which do not follow from it. The solver reproduces the exact exponents $\beta = 1/2$, $\gamma = 1$, $\delta = 3$.

The corpus does not identify the coupling, so no value is quoted

What was tried. Regress the outcome on the measured index and read off α .

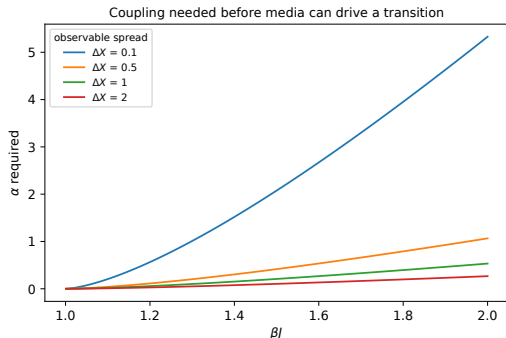
What happened. The fit performs *worse than the sample mean* out of sample, and the estimate scales inversely with a coupling βJ that no measurement here fixes.

So no value of $\hat{\alpha}$ appears anywhere — not in a table, a caption, or “for illustration”.

Reporting an unidentified coupling as though it were measured would be the failure here. Declining to quote it is not.

What replaces it is a statement that does not need the fit at all.

Any content observable must clear a bound to drive a transition



$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

Rather than fitting α and hoping it clears zero, this states what α *would have to be* for the mechanism to work at all — for **any** candidate observable, before a corpus exists.

Measured spread $\Delta X = 0.1241 \Rightarrow \alpha \geq 0.169$ at $\beta J = 1.10$, 1.655 at 1.50, **4.294** at 2.00

05

Standing

What holds, and what is still open.

Whether the encoder tracks real cortex is measurable, and one pass remains

A commercial lab reports the released checkpoint, in the averaged configuration this project runs, is **anti-correlated** with real cortex. Four subjects, one stimulus, not independently replicated.

0.1517

measured noise ceiling

CI [0.1484, 0.1551], 4
subjects, 1000 parcels

0.2247

withdrawn

an episode picked by position,
and a guard that could not fire

Replaying the old code path reproduces the wrong number *exactly*, which proved the cause. Both defects were silent: nothing crashed.

Schaefer et al. (2018); Algonauts 2025 (CC0); audit self-published, Sapient (2026).

Standing

Four claims the work supports, and four it does not

SUPPORTED

- A content observable that can be measured, with its reliability stated.
- That it separates these four corpora at $\eta^2 = 0.1068$, power attached.
- A bound any candidate observable must satisfy, usable before data exists.
- A measured noise ceiling for the checkpoint on public data.

NOT SUPPORTED

- That the index detects manipulation. Source is confounded; TF-IDF scores 0.9758.
- Anything about the amygdala. The checkpoint does not predict it.
- Any value of the coupling α . The corpus does not identify it.
- Any verdict on one article. 12.8% reverse between sessions.

The work yields a dissertation and three papers

78 pp

Dissertation

Complete, review answered.

12 pp

Paper 1

The bound and the screening criterion.
Physica A. Independent of the scan.

15 pp

Paper 2

The instrument, the corpus, the field bound. *Physica A*.

6 pp

Paper 3

The measured ceiling and the replication. *Imaging Neuroscience*. One computation on its final run.

Every number in all four comes from a re-runnable script. Nothing is transcribed by hand.

Conclusions

1. **A field observable can be measured from content.** The instrument runs, the corpus is complete, and reliability is measured rather than assumed: $ICC = 0.8725$.
2. **It separates four corpora at $\eta^2 = 0.1068$, and the separation is confounded with source.** Both halves are reported, the confound first.
3. **The coupling is unidentified, so the thesis reports a bound instead.** With $\Delta X = 0.1241$, $\alpha \geq 4.294$ at $\beta J = 2$ — a constraint on any future proposal of this mechanism.
4. **The instrument's own validity is now a measurable question.** The noise ceiling is 0.1517; one prediction pass remains.

References

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